Out of Eden

Tribal Migration Simulation — Numerical Mechanics Design Document

Version 2.0 — Complete Specification

0. Conventions & Notation

This document uses the following notation throughout:

- **p- :** proportional to / positively impacts the parameter.
- **n- :** inversely proportional to / negatively impacts the parameter.
- **clamp(x, lo, hi) :** constrains value x to the range [lo, hi].
- **lerp(a, b, t) :** linear interpolation: a + (b - a) * t.
- **rand() :** uniform random float in [0, 1).
- **randInt(a, b) :** uniform random integer in [a, b] inclusive.
- **year :** one simulation tick. All rates are per-tick unless stated otherwise.

All parameters use SI-adjacent units. Temperature in Celsius, population as integer headcount. Probability values range [0, 1]. Ability scores use a 0–100 scale unless stated otherwise.

1. Simulation Loop (Per Tick / Year)

The simulation processes each tick in a strict order to avoid race conditions and ensure deterministic reproducibility when given the same random seed. The order is:

- Phase 1: Global events — update global temperature drift, roll for global-scale events.
- Phase 2: Tile update — for each tile: natural recovery of carrying capacity, disaster rolls, plague propagation.
- Phase 3: Tribe update (order randomized each tick) — for each tribe:
    - 3a. Leader check (succession, abdication, death).
    - 3b. Compute pressures from tile state and tribe state.
    - 3c. Technology innovation rolls (driven by pressure).
    - 3d. Population dynamics (births, deaths, plague mortality).
    - 3e. Resource consumption and carrying capacity impact.
    - 3f. Ability decay for unused abilities.
    - 3g. Trade resolution with neighboring/co-located tribes.
    - 3h. Diplomacy update (alliance decay, relationship drift).
- Phase 4: Interaction phase — for each tile with multiple tribes:
    - 4a. Hostility rolls and combat resolution.
    - 4b. Cross-tile raids (neighboring tiles).
- Phase 5: Migration phase — for each tribe: migration decision and execution.
- Phase 6: Fission phase — for each tribe: cohesion check, potential split.
- Phase 7: Extinction check — remove tribes with population <= 0.

Tribe update order is randomized each tick to prevent systematic first-mover advantage. All writes within a phase use double-buffering: reads come from the start-of-phase snapshot, writes go to a staging buffer, and the buffer is committed before the next phase begins.

2. Global Parameters

These parameters set the baseline for the entire simulation. Each tribe may override some of these locally.

2.1 Birth Rate (G_birth)

Default: 0.035. Range: [0, 1]. Represents the base annual probability of each population unit producing a new individual. Modified at tribe level by food availability, comfort, and organization.

```
tribe_births = floor(pop * effective_birth_rate)
effective_birth_rate = G_birth * food_multiplier * comfort_multiplier *
org_multiplier
```

Where food_multiplier = clamp(1.0 - food_pressure * 0.8, 0.1, 1.2), comfort_multiplier = clamp(1.0 - (heat_pressure + cold_pressure) * 0.3, 0.5, 1.1), org_multiplier = 1.0 + organization_ability * 0.005.

2.2 Death Rate (G_death)

Default: 0.029. Range: [0, 1]. Base annual mortality per population unit. Modified by pressures.

```
tribe_deaths = floor(pop * effective_death_rate)
effective_death_rate = G_death + pressure_death_bonus
pressure_death_bonus = food_pressure * 0.04 + water_pressure * 0.05 +
cold_pressure * 0.03 + heat_pressure * 0.03
```

[NOTE] Net growth rate at defaults with zero pressure: 0.035 - 0.029 = 0.006 = 0.6% per year. This is historically plausible for pre-Neolithic populations, producing a doubling time of ~115 years.

2.3 Tribal Hostility (G_hostility)

Default: 0.2. Range: [0, 1]. Base probability per year that a tribe initiates an attack on any one other tribe sharing the same tile. Cross-tile raiding uses G_hostility * 0.5.

2.4 Disaster Frequency (G_disaster)

Default: 0.1. Range: [0, 1]. Per-tile, per-year probability of a disaster event being triggered. The disaster type is then selected based on tile properties (see Section 6).

2.5 Innovation Frequency (G_innovation)

Default: 0.002. Range: [-0.1, 1]. Per-tribe, per-year base probability of a technological advancement. Negative values represent entropy/regression tendency. This is HEAVILY modified at tribe level (see Section 7).

2.6 Cohesion (G_cohesion)

Default: 0.9. Range: [0, 1]. The probability that a tribe does NOT split in a given year. Hence, annual fission probability = 1 - G_cohesion = 0.1 at default. Modified by population and organization.

2.7 Global Mean Temperature (G_temp)

Default: 15°C. This parameter is necessary for long-span simulations: it models glacial/interglacial cycles and directly influences every tile's local temperature, which cascades into habitability, disaster types, and carrying capacity.

```
G_temp(year) = 15 + amplitude * sin(2 * PI * year / period + phase)
```

Where amplitude = 5 (modeling ~10°C glacial-interglacial swing), period = 100000 (years, Milankovitch cycle approximation). The user can also inject step-changes to model abrupt climate events.

```
tile_local_temp = tile_base_temp + (G_temp - 15)
```

2.8 Migration Tendency (G_migration)

Default: 0.1. Range: [0, 1]. Base annual probability that a tribe considers migration. Actual migration depends on pressure differentials (see Section 9).

3. Tile (地块) Parameters

The world is represented as a hex grid. Each tile has fixed geographic properties and derived dynamic properties. Tiles connect to 2–6 neighbors depending on position.

3.1 Climate Classification (Köppen)

Each tile is assigned a Köppen climate type that encodes both temperature and precipitation regime. This replaces separate temperature and rainfall parameters with a unified system.

Köppen Group	Examples	Base Temp °C	Annual Precip mm	Habitability
Af (Tropical Rainforest)	Congo Basin, SE Asia	26	2000+	4
Aw (Tropical Savanna)	East Africa, N Australia	25	800–1500	5
BSh (Hot Semi-Arid)	Sahel, Horn of Africa	28	250–500	3
BWh (Hot Desert)	Sahara, Arabian	30	< 250	1
Cfa (Humid Subtropical)	SE China, SE USA	18	1000–1500	4
Cfb (Oceanic)	W Europe, NZ	12	800–1200	5
Dfc (Subarctic)	Siberia, N Canada	-5	400–700	2
ET (Tundra)	Arctic coasts	-10	200–400	1
EF (Ice Cap)	Interior Greenland	-25	< 200	0

Habitability (0–5) is the composite comfort score. Base temperature is modified by G_temp drift: tile_effective_temp = base_temp + (G_temp - 15).

3.2 Terrain

Terrain	Movement Cost	Agriculture Modifier	Defense Bonus	Notes
Plains (平原)	1.0	1.2	0.0	Best farmland potential
Hills (丘陵)	1.5	0.8	+0.15	Decent farmland, defensive advantage
Plateau (高原)	1.3	0.6	+0.10	Altitude penalty to comfort
Mountains (高山)	3.0	0.1	+0.30	Nearly impassable, strong defense
Swamp (沼泽)	2.0	0.3	+0.05	Disease risk, poor farming
Coast (海岸)	1.0	0.9	0.0	Fishing bonus, trade hub potential
River Valley (河谷)	1.0	1.5	0.0	Best agriculture + water

3.3 Water Resources

Water availability is a composite of precipitation and surface water sources:

```
water_avail = precip_score + surface_water_bonus
```

Where precip_score maps from annual precipitation (0–5 scale: <250mm=0, 250–500=1, 500–800=2, 800–1200=3, 1200–2000=4, >2000=5), and surface_water_bonus adds +2 for major rivers, +1 for lakes/springs. Maximum water_avail is capped at 7.

3.4 Carrying Capacity

Each tile has dynamic carrying capacity divided into two components:

3.4.1 Foraging/Hunting Capacity (K_hunt)

Represents the maximum population sustainable through hunting-gathering-fishing.

```
K_hunt_base = habitability * 40 + water_avail * 15 + terrain_hunt_bonus
```

Where terrain_hunt_bonus: coast +50 (fishing), river_valley +30, swamp +15, plains +10, hills +5, plateau 0, mountains -10.

Depletion mechanic: If local population relying on foraging exceeds 85% of K_hunt, capacity begins to decline:

```
overshoot_ratio = max(0, foraging_pop / K_hunt - 0.85)
K_hunt_next = K_hunt * (1 - overshoot_ratio * 0.67) + recovery
recovery = min(K_hunt_base * 0.05, K_hunt_base - K_hunt)
```

Recovery is capped so that K_hunt never exceeds K_hunt_base. If foraging_pop drops below 85% of K_hunt, recovery occurs at 5% of base per year.

3.4.2 Agricultural Capacity (K_agri)

Initially 0 for all tiles. Activated only when a tribe with agriculture ability occupies the tile.

```
K_agri_base_potential = terrain_agri_mod * water_avail * 80
K_agri_growth_rate = 0.003 * pop_farming * agri_ability / 100 *
terrain_agri_mod
K_agri_next = min(K_agri + K_agri_growth_rate, K_agri_base_potential)
```

Agricultural capacity grows slowly (representing land clearing). It does not naturally decay but can be damaged by floods, drought, and neglect (if no tribe with agriculture occupies the tile for 20+ years, K_agri decays at 2% per year).

> *[NOTE] The growth rate is deliberately much slower than population natural increase (0.6%). A tribe hitting its foraging ceiling must innovate agriculture AND wait for fields to be developed — creating a realistic population pressure squeeze.*

3.4.3 Water Carrying Capacity (K_water)

```
K_water = water_avail * 100 + water_engineering_bonus
water_engineering_bonus = sum(tribe.water_eng_ability * tribe.pop * 0.01)
for all tribes on tile
```

If total population on tile exceeds K_water, a water stress penalty applies to all tribes: death rate increases and birth rate decreases proportionally.

3.5 Tile Comfort (Derived)

```
comfort = habitability - altitude_penalty - overcrowding_penalty
```

Where altitude_penalty = 1 for plateau, 2 for mountains; overcrowding_penalty = max(0, total_pop / total_K - 1.0) * 2.

4. Tribe (部落) Parameters

Each tribe is a distinct entity with its own state. When a tribe overrides a global parameter, the override is absolute (not additive).

4.1 Population

Integer. The fundamental resource. Population directly or indirectly influences almost every other system.

```
pop_next = pop + births - deaths - combat_losses - emigration + immigration
```

4.2 Birth Rate & Death Rate

See Section 2.1 and 2.2 for base formulas. Tribe-specific overrides replace G_birth / G_death directly.

4.3 Tribal Hostility

Default: inherits G_hostility. Can be overridden. Modified by leader trait (Conqueror: *3), competition pressure, and relationship scores with specific tribes.

```
effective_hostility(target) = base_hostility * leader_mod * (1 +
competition_pressure * 0.5) * relationship_hostility_mod(target)
```

4.4 Leader System

Each tribe has at most one leader. Leaders are generated randomly and provide significant modifiers.

4.4.1 Leader Generation

When a tribe has no leader (initial state or after departure), a new leader emerges with probability 0.3 per year. Leader properties:

- **Competence:** randInt(-25, 25). Applies as a percentage modifier to ALL ability checks: ability * (1 + competence/100).
- **Trait:** One of the following, selected uniformly at random:
  - Conqueror (征服者): tribal hostility * 3, attack ability +15%.
  - Innovator (创新者): innovation rate * 5, all ability gain +10%.
  - Explorer (探索者): migration tendency * 3, scouting range +1 tile.
  - Organizer (组织者): organization ability base +25%, cohesion +10%.
  - Diplomat (外交家): relationship improvement rate * 2, trade bonus +20%.
  - Sage (贤者): all ability decay halved, tech innovation * 2 (stacks with Innovator differently — Sage slows loss; Innovator accelerates gain).

4.4.2 Leader Departure

```
departure_prob = 0.05 + max(0, age_in_office - 20) * 0.02
```

A leader departs (death, overthrow, retirement) each year with the above probability. Negative-competence leaders have +0.02 additional departure probability per year (unpopular). When a Conqueror leader departs, there is a 30% chance of a succession crisis: the tribe loses 10% population and cohesion drops by 0.15 temporarily.

4.5 Abilities (0–100 Scale)

Each ability has a cap (maximum achieved level) and a current (effective) value. Current can temporarily drop below cap due to disasters or events, and recovers toward cap at 5% of the gap per year.

```
current_next = current + (cap - current) * 0.05
```

4.5.1 Foraging Ability (渔猎能力)

```
effective_foraging = pop * 0.5 * (1 + foraging_ability_current / 100)
```

This determines how much of K_hunt the tribe can actually exploit. If effective_foraging < tribe_pop, the shortfall creates food pressure.

Cap growth: primarily population-driven. Innovation adds percentage enhancement.

```
foraging_cap_growth = innovation_roll ? min(5, pressure_driven_gain) : 0
```

4.5.2 Agriculture Ability (农业能力)

Starts at 0. Cannot be gained until the historical agriculture precondition system activates (see Section 7). Once unlocked:

```
effective_farming = pop * 0.3 * (1 + agri_ability_current / 100) *
terrain_agri_mod
```

Cap growth: primarily innovation-driven. Population provides minor bonus.

4.5.3 Heat Tolerance (耐热能力)

Population-independent. Gained only through random innovation when heat_pressure > 0.

```
heat_tolerance_gain_prob = G_innovation * (1 + heat_pressure * 10)
```

4.5.4 Cold Tolerance (耐寒能力)

Gained through innovation when cold_pressure > 0. Innovation and population both contribute.

```
cold_tolerance_gain_prob = G_innovation * (1 + cold_pressure * 8) * (1 +
pop / 500 * 0.1)
```

4.5.5 Water Engineering (水利工程)

Depends on population, innovation, and organization. Provides bonus to tile water capacity.

```
water_eng_effective = water_eng_cap * (1 + organization / 100 * 0.5) * (1 +
pop / 200 * 0.1)
```

4.5.6 Attack Ability (攻击能力)

```
combat_power = pop * (1 + attack_ability / 100) * terrain_defense_mod * (1
+ org / 100 * 0.3)
```

Positively influenced by population, innovation-driven ability gains, and organization.

4.5.7 Organization (组织能力)

Starts at 0. Population-independent base growth through innovation. High organization mitigates the negative effect of population on cohesion, improves combat, trade, and birth rate.

```
org_cohesion_bonus = organization / 100 * 0.15
```

This means max organization (100) provides +15% cohesion, counteracting population-driven fission.

4.5.8 Ability Decay

For every ability where the tribe currently experiences NO corresponding pressure, each year there is a 1% probability that the ability cap decreases by 1 point. Sage leaders halve this to 0.5%.

```
if (rand() < 0.01 * sage_mod && pressure_for_ability == 0) { ability_cap -= 1; }
```

4.6 Cohesion & Fission

```
effective_cohesion = G_cohesion - pop_penalty + org_cohesion_bonus + leader_bonus
pop_penalty = clamp(log2(pop / 150) * 0.05, 0, 0.4)
```

Fission probability = 1 - effective_cohesion. When fission occurs:

- Population splits: each child tribe gets 40–60% (random).
- Abilities: both tribes inherit parent ability caps with -5 penalty (representing knowledge loss during social disruption).
- The child tribes have a neutral-to-positive relationship with each other (kinship bonus: +0.3 relationship score).
- Leaders: one child inherits the leader (random); the other starts leaderless.

5. Pressure System

Pressures are derived values (0–1 scale) that drive innovation, migration, conflict, and mortality. They are the engine of emergence: pressure creates innovation, innovation relieves pressure, but population growth re-creates pressure.

5.1 Food Pressure

```
total_food_capacity = effective_foraging + effective_farming
food_pressure = clamp(1 - total_food_capacity / pop, 0, 1)
```

5.2 Heat Pressure

```
heat_pressure = clamp((tile_effective_temp - 22 - heat_tolerance * 0.3) /
20, 0, 1)
```

5.3 Cold Pressure

```
cold_pressure = clamp((5 - tile_effective_temp + cold_tolerance * 0.3) /
25, 0, 1)
```

5.4 Water Pressure

```
water_pressure = clamp(1 - K_water / (pop * 0.8), 0, 1)
```

5.5 Competition Pressure

```
competition_pressure = clamp(any_other_pressure * co_tile_tribes_factor, 0,
1)
co_tile_tribes_factor = (num_tribes_on_tile - 1) * 0.3 + neighbor_tribes *
0.1
```

Competition pressure emerges when a tribe has any existing pressure AND shares space with other tribes. This creates arms races: pressure motivates conflict, which motivates attack ability, which motivates defensive innovation in neighbors.

5.6 Organization Pressure

```
org_pressure = clamp(log2(pop / 100) * 0.15 - organization / 100 * 0.5, 0,
1)
```

Larger populations inherently need more organization. Without it, cohesion drops and the tribe fragments.

5.7 Total Pressure (for innovation weighting)

```
total_pressure = food + heat + cold + water + competition + org
```

When an innovation roll succeeds, the specific ability improved is chosen weighted by the relative magnitude of each pressure component.

6. Disaster System

Each tick, each tile rolls for disaster occurrence. The probability and type are shaped by tile properties.

```
disaster_roll = rand() < G_disaster * tile_disaster_modifier
```

6.1 Disaster Types

6.1.1 Wildfire (火灾)

Conditions (p-): forested Köppen types (Af, Aw, Cfa, Cfb), low humidity, low water, high temperature.

```
fire_weight = forest_factor * (1 - water_avail/7) * clamp((temp - 20)/15,
0, 1)
```

Effects: K_hunt * 0.4 (severe drop), all tribes on tile: foraging_current * 0.6, agri_current * 0.9, tile comfort -2 for 3 years.

6.1.2 Flood (洪灾)

Conditions (p-): high precipitation, rivers, swamp, plains.

```
flood_weight = (precip_score / 5) * (terrain == swamp ? 1.5 : terrain ==
plains ? 1.2 : terrain == river_valley ? 1.3 : 0.5)
```

Effects: K_hunt * 0.8, all tribes: agri_current * 0.4 (devastating to farms), K_agri * 0.85, comfort -1 for 2 years.

6.1.3 Natural Plague (自然瘟疫)

Occurs randomly. Upon occurrence, generates a plague entity with:

- Contagion: uniform(0.05, 0.5) — probability of spreading to each adjacent tile per year.
- Duration: randInt(1, 5) years per affected tile.
- Severity: uniform(0.3, 0.8) — multiplier to K_hunt reduction.

Propagation: each year, an affected tile spreads to each neighbor with probability = contagion. Each subsequent hop: contagion *= 0.5. Below 0.05, no further spread. Different tiles may have different remaining durations.

Effects on each affected tile: K_hunt * (1 - severity * 0.5), all tribes: foraging_current * (1 - severity * 0.3), agri_current * (1 - severity * 0.2).

6.1.4 Severe Winter (严冬)

Conditions (p-): Köppen cold/temperate types (Dfc, ET, Cfb), high altitude.

```
winter_weight = cold_factor * (1 + altitude_factor * 0.5)
```

Effects: K_hunt * 0.3, all tribes: foraging_current * 0.4, agri_current * 0.5, K_agri * 0.95, comfort -3 for 1 year.

6.1.5 Drought (干旱)

Conditions (p-): low precipitation, arid/semi-arid Köppen.

```
drought_weight = (1 - precip_score / 5) * arid_factor
```

Effects: water_avail temporarily * 0.3, K_hunt * 0.4, K_agri * 0.8, all tribes: agri_current * 0.3. Duration: 1–3 years.

6.1.6 Volcanic Event (火山事件) [NEW]

Only on tiles marked volcanic. Rare (weight * 0.1). Effects: massive K_hunt and K_agri reduction on tile AND adjacent tiles. Comfort -4 for 5 years. But: soil fertility +20% to K_agri_base_potential after recovery (volcanic soil).

6.1.7 Earthquake (地震) [NEW]

Only on tiles near tectonic boundaries. Effects: K_agri * 0.7, organization_current * 0.8 for all tribes (infrastructure damage), comfort -1 for 1 year.

6.2 Disaster Type Selection

When a disaster triggers, the type is selected probabilistically based on relative weights. Weights are computed from tile properties. If all weights are near zero, the disaster fizzles (no effect). This means deserts rarely flood, and tundra rarely burns.

7. Technology & Innovation System

7.1 Innovation Roll

```
innovation_prob = G_innovation * pop_factor * pressure_factor *
neighbor_factor * leader_mod
pop_factor = clamp(log10(pop) / 3, 0.1, 2.0)
pressure_factor = 1 + total_pressure * 5
neighbor_factor = 1 + sum(neighbor_tribe_max_ability / 100) * 0.1 for each
neighbor tribe
leader_mod = innovator ? 5.0 : sage ? 2.0 : 1.0
```

Each tick, the tribe rolls rand() < innovation_prob. On success, one ability gains +1 to +5 cap points (amount = 1 + floor(pressure_for_that_ability * 4)). The ability improved is selected weighted by pressure magnitudes.

7.2 Technology Diffusion

Tribes on the same tile or adjacent tiles with higher ability scores create a diffusion pull. Each tick, for each ability:

```
if any neighbor has ability_cap > my_ability_cap + 10:
  diffusion_prob = 0.02 * (best_neighbor_cap - my_cap) / 100 * trade_bonus
  if rand() < diffusion_prob: my_cap += randInt(1, 3)
```

Trade relationships amplify diffusion (see Section 10). Hostile relationships reduce it to 0.

7.3 Agriculture Evolution Timeline [NEW]

Agriculture does not appear as a sudden invention. It follows a staged progression:

Stage 0: Pure Foraging

Initial state. Tribes rely entirely on K_hunt.

Stage 1: Intensified Gathering (强化采集)

Prereq: food_pressure > 0.3 for 5+ consecutive years AND pop > 80. Effect: foraging_ability gains +10 bonus (representing selective harvesting, better tools). agri_ability remains 0.

```
gathering_unlock_prob = 0.01 * (food_pressure - 0.3) per qualifying year
```

Stage 2: Proto-Agriculture (前农业)

Prereq: Stage 1 achieved AND food_pressure > 0.2 for 10+ years AND pop > 150 AND gathering_ability_cap > 30. Effect: agri_ability_cap set to 5. K_agri begins at 10% of potential.

```
proto_agri_prob = 0.005 * (food_pressure) * pop_factor per qualifying year
```

Stage 3: Full Agriculture (农业)

Prereq: Stage 2 achieved AND agri_ability_cap > 20 AND sustained farming population. Effect: normal agriculture mechanics fully active, innovation can push agri_ability without cap.

> *[NOTE] This staged system means agriculture typically cannot appear before ~10,000 BC in a standard simulation starting at ~70,000 BC, matching the archaeological record. However, it CAN emerge earlier under extreme pressure — an emergent property, not a hard-coded timeline.*

8. Combat System

8.1 Hostility Check

Each tribe on a tile checks for aggression against every other tribe it can see:

- Same-tile tribes: attack_prob = effective_hostility(target).
- Adjacent-tile tribes: attack_prob = effective_hostility(target) * 0.5 / terrain_cost_of_target_tile.

If multiple tribes on the same tile are hostile, each pair resolves independently. This naturally creates chaotic multi-front conflicts.

8.2 Combat Resolution

```
attacker_power = attacker.pop * (1 + attacker.attack_ability / 100) * (1 +
attacker.org / 100 * 0.3)
defender_power = defender.pop * (1 + defender.attack_ability / 100) * (1 +
defender.org / 100 * 0.3) * (1 + terrain_defense)
power_ratio = attacker_power / (attacker_power + defender_power)
attacker_losses = floor(attacker.pop * 0.1 * (1 - power_ratio) * intensity)
defender_losses = floor(defender.pop * 0.1 * power_ratio * intensity)
```

Where intensity = uniform(0.5, 1.5), representing battle scale. In a decisive victory (power_ratio > 0.7), the loser may be forced to migrate (30% chance) or absorbed (10% chance if ratio > 0.85).

8.3 Siege & Alliance Warfare [NEW]

Tribes in an alliance can coordinate attacks. When an alliance member declares war, allies with hostility toward the same target have a 50% chance of joining. Allied combat power is summed but with a coordination penalty:

```
allied_power = sum(individual_powers) * (0.7 + 0.3 * average_org / 100)
```

Low-organization alliances fight poorly together. This creates pressure for organization development in warlike alliances.

9. Migration System

9.1 Migration Decision

```
migrate_prob = G_migration * (1 + total_pressure * 3) * leader_mod * (1 - comfort_retention)
comfort_retention = clamp(tile_comfort / 5 * 0.5, 0, 0.5)
```

Comfortable, low-pressure tribes rarely migrate. High-pressure tribes with Explorer leaders migrate frequently.

9.2 Destination Selection

When migration triggers, the tribe evaluates all accessible neighboring tiles:

```
tile_score = habitability * 2 + water_avail + (K_hunt - current_foraging_pop) / 100 - num_hostile_tribes * 2 - terrain_cost
```

The tribe selects the highest-scoring tile probabilistically (softmax with temperature=0.5 to allow some randomness). If all neighbors score worse than current tile, there is a 50% chance the tribe stays. Explorer leaders reduce this chance to 20%.

9.3 Partial Migration [NEW]

Instead of the entire tribe always migrating, large tribes (pop > 200) can send a splinter group:

```
splinter_prob = 0.3 if pop > 200 AND migration triggered
```

If a splinter migration occurs, 20–40% of the population departs as a new tribe, inheriting abilities at -10 cap penalty and starting with no leader. This models colonization without abandoning the homeland.

10. Trade System

Trade is an emergent interaction between tribes, not a resource the player directly controls. It arises from proximity and non-hostility.

10.1 Trade Eligibility

```
can_trade(A, B) = same_tile(A, B) || adjacent_tile(A, B)
  AND relationship(A, B) > -0.3  // not actively hostile
  AND both populations > 20
```

10.2 Trade Value

```
trade_value(A, B) = ability_difference_bonus + pop_complementarity
ability_difference_bonus = sum(|A.ability[i] - B.ability[i]|) /
(num_abilities * 100)
pop_complementarity = min(A.pop, B.pop) / max(A.pop, B.pop) * 0.5
```

Tribes with different specializations trade more valuably. Extremely asymmetric populations trade less effectively.

10.3 Trade Effects

- Both tribes: birth_rate * (1 + trade_value * 0.1) — prosperity bonus.
- Both tribes: innovation rate * (1 + trade_value * 0.3) — cross-pollination.
- Technology diffusion rate * (1 + trade_value * 2.0) — accelerated learning.
- Relationship(A, B) improves by 0.02 * trade_value per year.
- Coast tiles and river_valley tiles: trade_value * 1.5 (trade hub bonus).

[NOTE] Trade creates positive feedback: trading improves relationships, which improves trade, which accelerates innovation. But it also attracts raiders (wealthy tiles become targets for hostile tribes). This tension drives complex emergent dynamics.

11. Diplomacy & Alliance System

11.1 Relationship Score

Each tribe-pair has a relationship score in [-1, 1]. Initial score: 0 (neutral). Score shifts:

- +0.02/year for co-tile coexistence without conflict.
- +0.3 initial bonus for fission sibling tribes (kinship).
- -0.3 per combat event between the tribes.
- +0.02 * trade_value per year of active trade.
- -0.01/year natural decay toward 0 (relationships fade without reinforcement).

11.2 Alliance Formation

```
alliance_prob = max(0, relationship - 0.5) * 0.1 * diplomat_mod per year
```

Alliances provide: shared defense (attacked ally triggers mutual defense), trade bonus *1.5, tech diffusion *2. Alliance members cannot attack each other (hostility checks auto-fail).

11.3 Alliance Dissolution

If relationship drops below 0.2 (due to resource competition, leader change, etc.), the alliance dissolves. Betrayal (attacking an ally) immediately sets relationship to -0.8 and dissolves the alliance.

11.4 Multi-Tribe Tile Tension [NEW]

When multiple tribes share a tile, tension scales non-linearly:

```
tile_tension = (num_tribes - 1)^1.5 * (1 - average_pairwise_relationship) *
overcrowding_factor
overcrowding_factor = clamp(total_pop / total_K - 0.5, 0, 2)
```

High tension increases all hostility checks on the tile. This means peaceful coexistence is possible on abundant tiles with good relationships, but scarce tiles naturally drive conflict.

12. Human Plague System

Distinct from natural plagues (which affect the tile). Human plagues affect tribes directly.

12.1 Outbreak

```
plague_outbreak_prob = base * pop_factor * climate_factor
base = 0.001
pop_factor = clamp(pop / 500, 0.1, 3.0)
climate_factor = tropical: 2.0, subtropical: 1.5, temperate: 1.0, cold: 0.5
```

Larger, denser populations in warm climates breed disease more readily.

12.2 Plague Properties

- Lethality: uniform(0.05, 0.3) — additional death rate per year while active.
- Contagion: uniform(0.05, 0.5) — probability of spreading per adjacent tribe per year.
- Duration: randInt(1, 5) years.

Propagation uses the same hop-decay mechanic as natural plagues (contagion * 0.5 per hop).

12.3 Effects

```
death_rate += lethality
cohesion -= lethality * 0.3  // social disruption
organization_current *= (1 - lethality * 0.2)  // institutional damage
```

13. Emergence Mechanisms

The following feedback loops are deliberately designed to produce emergent complex behavior. None of these outcomes are hard-coded; they arise naturally from the interaction of systems.

13.1 Positive Feedback Loops

- **Population-Innovation Spiral:** More people → more innovation → higher carrying capacity → more people. Checked by: resource depletion, disease, fission.
- **Trade-Prosperity Spiral:** Trade → innovation → specialization → more trade value. Checked by: raider attraction, overcrowding tension.
- **Military Arms Race:** Competition pressure → attack ability innovation → neighbors feel more pressure → their attack innovation. Checked by: combat losses, fragmentation.
- **Organization-Stability Spiral:** High org → high cohesion → large pop → more org pressure → more org innovation. Checked by: leader departure, plague.

13.2 Negative Feedback Loops

- **Malthusian Trap:** Pop growth → resource pressure → higher death rate + lower birth rate + migration. The classic brake on growth.
- **Empire Fragility:** Large pop → low cohesion → fission. Counter: organization. But organization takes time, creating a race condition.
- **Disaster Reset:** High population on a tile → overexploitation → fire/drought/famine risk. Reduces pop, allowing ecosystem recovery.
- **Plague Density Penalty:** Dense populations breed plagues → death → reduced density. Parallels real historical dynamics.

13.3 Expected Emergent Phenomena

- **Boom-Bust Cycles:** Tribes grow, hit carrying capacity, fragment or suffer disaster, then regrow. Irregular oscillation.
- **Technology Races:** Neighboring tribes develop in parallel but different directions (one military, one agricultural), creating geopolitical dynamics.
- **Migration Waves:** A disaster or overpopulation in one region creates a cascade of migrations as displaced tribes push into occupied territory.
- **Cultural Zones:** Trade networks create clusters of high-ability tribes that share technology, surrounded by lower-tech peripheries.
- **Agricultural Revolution:** Under the right pressure conditions, agriculture emerges and creates explosive population growth, which then creates new pressures (organization, plague, conflict).
- **Rise and Fall of Empires:** A Conqueror leader + high attack + high organization = rapid expansion. Leader departure + succession crisis + overextension = fragmentation. Not scripted; emergent.

14. Implementation Notes

14.1 Data Structures

```
World { tiles: Map<TileID, Tile>, tribes: Map<TribeID, Tribe>, year: int,
G_*: globals }
Tile { id, neighbors: TileID[], climate: Koppen, terrain, water_avail,
K_hunt, K_agri, ... }
Tribe { id, tile: TileID, pop, abilities: {cap, current}[], leader?,
relationships: Map<TribeID, float>, ... }
```

14.2 Random Seed

All randomness derives from a seeded PRNG (e.g., Mersenne Twister or xorshift128+). Given the same seed and parameters, the simulation is fully deterministic. This is critical for reproducibility and debugging.

14.3 Performance Considerations

- Double-buffered state prevents order-dependent bugs.
- Tribe update order randomized per tick prevents systematic advantage.
- Plague propagation uses BFS from infected tiles, not O(n^2) pair checks.
- Relationship storage: sparse map (only store non-zero relationships).
- For large worlds (1000+ tiles), tile-based spatial indexing for neighbor lookups.

14.4 Parameter Sensitivity

The following parameters most strongly affect simulation character:

- G_innovation: too high = instant tech utopia; too low = permanent stagnation. Default 0.002 is calibrated for ~70,000 year runs.
- G_cohesion: too low = tribes never grow; too high = single mega-tribe dominates.
- G_hostility: too low = no conflict; too high = extinction cascades.
- K_hunt recovery rate (5%): too fast = no resource pressure; too slow = permanent depletion.

14.5 Map Generation Suggestions

For an Africa-origin simulation, the map should include:

- East African Rift Valley tiles (Aw climate, river_valley terrain, high K_hunt) as starting region.
- Sahara barrier (BWh desert tiles) requiring cold tolerance or coastal route to cross.
- Levant corridor (BSh/Cfa) as gateway to Eurasia.
- A variety of climate zones to create diverse pressure environments.
- Coastal routes that provide consistent (if low) carrying capacity for migration paths.
- Mountain barriers (Himalayas, Alps analogs) that channelize movement.

15. Potential Future Extensions

- Domestication of animals: parallel track to agriculture, providing food + transport + military bonus.
- Language/culture system: tribes that share ancestry have cultural similarity, enabling easier diplomacy. Divergence over time increases barriers.
- Sea travel: unlockable ability allowing coastal-to-coastal migration, bypassing land barriers.
- Religion/ideology: emergent belief systems that provide cohesion bonus but create inter-group hostility with different ideologies.
- Metal working: late-game tech that dramatically boosts agriculture (plows) and combat (weapons).
- Climate-triggered mega-events: ice ages that render northern tiles uninhabitable, forcing southward migration cascades.
- Writing: unlockable at high organization + agriculture, dramatically reducing ability decay and boosting tech diffusion.
- Specialized occupations: at high enough organization, tribes develop internal specialization, improving all abilities but increasing fragility to disruption.

Appendix A: Global Parameter Quick Reference

Parameter	Default	Range	Primary Effect
G_birth	0.035	[0, 1]	Base births per pop per year
G_death	0.029	[0, 1]	Base deaths per pop per year
G_hostility	0.2	[0, 1]	Base attack probability
G_disaster	0.1	[0, 1]	Per-tile disaster probability
G_innovation	0.002	[-0.1, 1]	Per-tribe innovation probability
G_cohesion	0.9	[0, 1]	1 - fission probability
G_temp	15°C	[-25, 35]	Global mean temperature
G_migration	0.1	[0, 1]	Base migration probability

Appendix B: Ability Quick Reference

Ability	Scale	Primary Driver	Counters Pressure	Decay?
Foraging (渔猎)	0-100	Population + Innovation	Food	Yes (1%/yr)
Agriculture (农业)	0-100	Innovation (staged unlock)	Food	Yes (1%/yr)
Heat Tolerance (耐热)	0-100	Random + Innovation	Heat	Yes (1%/yr)
Cold Tolerance (耐寒)	0-100	Innovation + Pop minor	Cold	Yes (1%/yr)
Water Eng. (水利)	0-100	Pop + Innovation + Org	Water	Yes (1%/yr)
Attack (攻击)	0-100	Pop + Innovation	Competition	Yes (1%/yr)
Organization (组织)	0-100	Innovation only	Org pressure	Yes (1%/yr)

Appendix C: Disaster Quick Reference

Disaster	Primary Condition	K_hunt Impact	K_agri Impact	Duration
Wildfire	Forest + dry + hot	* 0.4	* 0.9 (current)	3 yr comfort
Flood	High precip + low terrain	* 0.8	* 0.85 + current*0.4	2 yr comfort
Natural Plague	Random	* (1-sev*0.5)	current * (1-sev*0.2)	1-5 yr
Severe Winter	Cold climate + altitude	* 0.3	* 0.95 + current*0.5	1 yr comfort
Drought	Low precip + arid	* 0.4	* 0.8 + current*0.3	1-3 yr

Volcanic	Volcanic tile (rare)	* 0.1	* 0.1 then +20% base	5 yr comfort
Earthquake	Tectonic boundary	unchanged	* 0.7	1 yr comfort